NETWORK INFRASTRUCTURE PROJECT DOCUMENTATION

Multi-Router Network with OSPF, DHCP, DNS, Web, Email & FTP Services

Cisco Packet Tracer Project

Author: Patrik

March 2026

Table of Contents

Contents

1. Project Overview 5
1.1 Technologies Used 5
1.2 Network Topology Overview 5
2. IP Addressing and Subnetting 7
2.1 Understanding Subnetting 7
2.1.1 The Subnetting Formula 7
2.1.2 Our Subnetting Approach 7
2.1.3 Block Size Calculation 7
2.1.4 Detailed Subnet Breakdown — Subnet 1 8
2.1.5 Detailed Subnet Breakdown — Subnet 2 8
2.1.6 Expanding Beyond Two Subnets 9
2.2 Complete Subnet Allocation Table 9
3. Device Inventory 10
3.1 Routers 10
3.2 Switches 10
3.3 End Devices 10
4. DHCP Configuration 12
4.1 Overview 12
4.2 DHCP Pool Summary 12
4.3 MainRouter DHCP Configuration (CLI) 12
4.4 Router2 DHCP Configuration (CLI) 13
4.5 Verifying DHCP 13
5. OSPF Dynamic Routing 14
5.1 Overview 14
5.2 OSPF Network Advertisements 14
5.2.1 Understanding Wildcard Masks 14
5.3 MainRouter OSPF Configuration (CLI) 14
5.4 Router2 OSPF Configuration (CLI) 15
5.5 Verifying OSPF 15
6. Network Security 17
6.1 Overview 17
6.2 Security Configuration Summary 17
6.3 MainRouter Security Configuration (CLI) 17

6.4 Switch Security Configuration Example (CLI) 18
6.5 Switch Management IP Addresses 18
6.6 Testing SSH Access 18
7. VLANs (Virtual LANs) 19
7.1 Overview 19
7.2 VLAN Configuration (CLI) 19
7.3 VLAN Verification and Testing 19
8. Wireless Network Configuration 21
8.1 Overview 21
8.2 Physical Module Installation 21
8.3 WirelessRouter Configuration 22
8.4 Laptop Configuration 22
9. Server Services 24
9.1 Server Overview 24
9.2 DNS Configuration 24
9.3 Web Server (HTTP) 24
9.4 Email Server (SMTP/POP3) 25
9.4.1 Email User Accounts 25
9.4.2 Email Client Configuration on PCs 25
9.5 FTP Server 27
9.5.1 FTP User Accounts and Permissions 27
9.5.2 FTP Permission Legend 28
9.5.3 FTP Testing 28
10. Inter-Router Connectivity 29
10.1 Point-to-Point Link 29
10.2 Configuration (CLI) 29
10.3 Router2 Physical Module 29
11. Configuration Management 30
11.1 Saving Configurations 30
12. Network Testing and Verification 31
12.1 Connectivity Tests 31
12.2 Service Tests 31
12.3 Complete Traffic Flow Example 31
13. Lessons Learned and Troubleshooting 33
13.1 Key Lessons 33
13.2 NAT Attempt 33

14. Future Enhancements 34
15. Verification Commands Reference 34
15.1 Overview 34
15.2 Router Verification Commands 34
15.2.1 show ip interface brief 34
15.2.2 show ip route 35
15.2.3 show ip ospf neighbor 37
15.2.4 show ip ospf interface 37
15.2.5 show ip dhcp pool 37
15.2.6 show ip dhcp binding 39
15.2.7 show running-config 39
15.3 Switch Verification Commands 40
15.3.1 show vlan brief 40
15.3.2 show interfaces trunk 41
15.3.3 show interfaces status 41
15.4 PC Verification Commands 41
15.4.1 ping 41
15.4.2 ipconfig 42
15.4.3 ftp 43
15.4.4 ssh 43
16. Packet Tracer Screenshots Guide 43
16.1 Topology Screenshots 43
16.2 Service GUI Screenshots 44
16.3 Physical Module Screenshots 44
16.4 Wireless Configuration Screenshots 44
16.5 Security Screenshots 45

1. Project Overview

This document provides comprehensive documentation for a multi-site enterprise network designed and implemented in Cisco Packet Tracer. The network simulates a real-world corporate infrastructure for Copa.hr, featuring five subnets across two routers, dynamic routing with OSPF, automatic IP addressing via DHCP, centralized services including DNS, Web, Email, and FTP, wireless access, and device security with SSH.

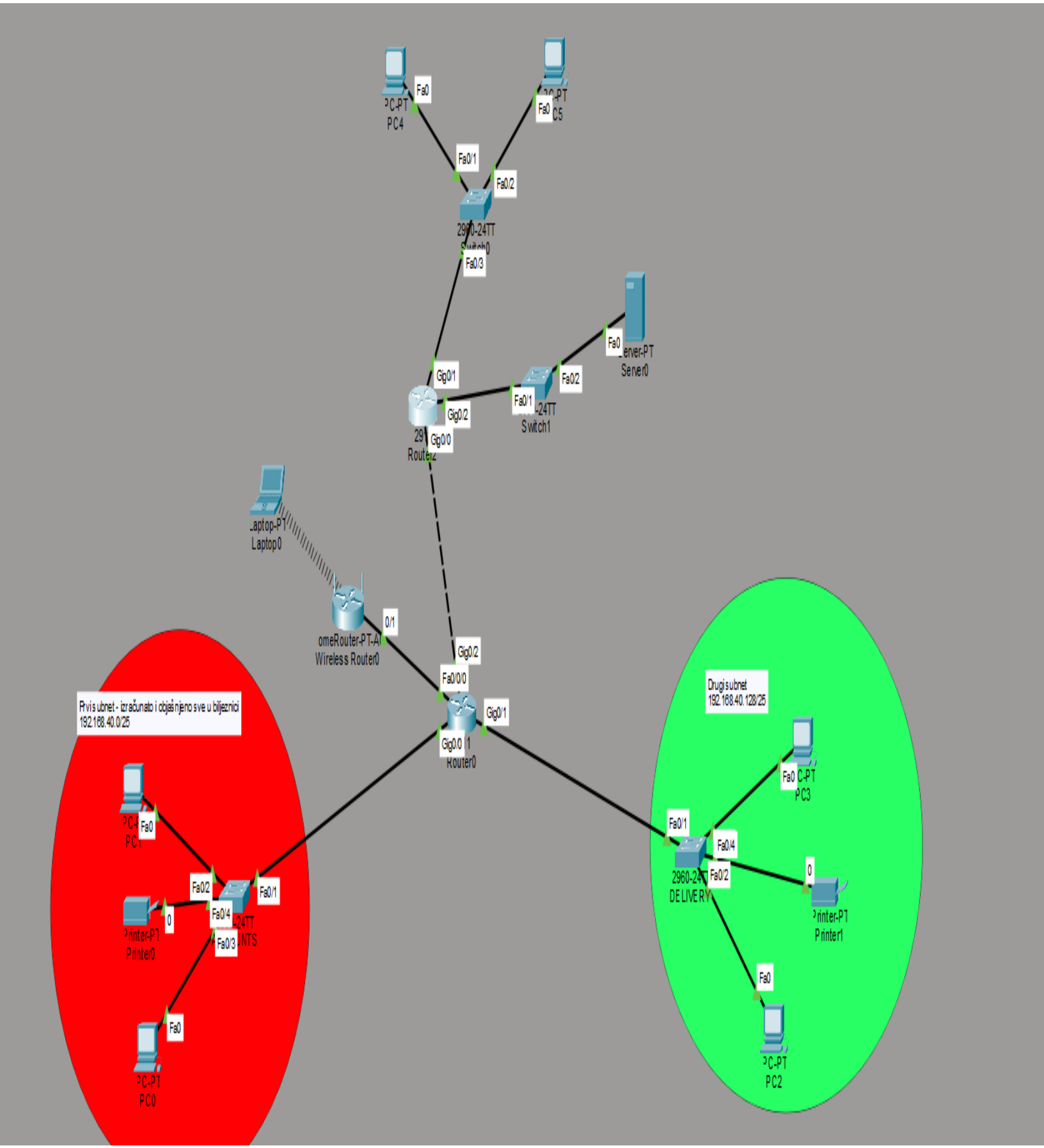
1.1 Technologies Used

The following technologies and protocols were implemented throughout this project:

Technology	Purpose	Where Applied
Subnetting (VLSM)	Divide network into logical segments	All 5 subnets + router links
DHCP	Automatic IP address assignment	All PC subnets + wireless
OSPF (Area 0)	Dynamic routing between routers	MainRouter and Router2
DNS	Domain name resolution	Server (copa.hr)
HTTP	Company website hosting	Server (www.copa.hr)
SMTP/POP3	Email communication	Server (copa.hr email)
FTP	File sharing with permissions	Server (role-based access)
SSH	Secure remote device management	All routers and switches
VLANs	Traffic segmentation (learned)	Both switches (demonstration)
NAT	Private to public IP translation (learned)	Conceptual understanding
Wireless (WiFi)	Wireless laptop connectivity	WirelessRouter + Laptop

1.2 Network Topology Overview

The network consists of two core routers (MainRouter and Router2) connected via a point-to-point /30 link. MainRouter serves Subnet 1 (Accounts), Subnet 2 (Delivery), and a wireless network (Subnet 4). Router2 serves Subnet 3 (general users) and the server subnet (Subnet 5 — server farm). An ISP router was planned for NAT configuration but was not implemented due to hardware module limitations in Packet Tracer.


PC-PT
PC4
Fa0
Fa0/1
Fa0/2
2960-24TT
Switch0
Fa0/3
Server-PT
Server0
Fa0
Fa0/2
Fa0/1
-24TT
Switch1
Gig0/1
Gig0/2
Gig0/0
Laptop-PT
Laptop0
0/1
Wireless Router0
Gig0/2
Fa0/0/0
Gig0/1
Gig0/0
Router0
Prvi subnet - izračunato i objašnjeno sve u bilježnici
192.168.40.0/25
Fa0
Fa0/2
Fa0/1
Fa0/4
Fa0/3
0
Printer-PT
Printer0
Fa0
PC-PT
PC0
Drugi subnet
192.168.40.128/25
Fa0
PC-PT
PC3
Fa0/1
Fa0/4
Fa0/2
2960-24
DELIVERY
0
Printer-PT
Printer1
Fa0
PC-PT
PC2

2. IP Addressing and Subnetting

2.1 Understanding Subnetting

Subnetting is the process of dividing a large network into smaller, more manageable sub-networks (subnets). This is essential for organizing network traffic, improving security, and making efficient use of IP addresses.

2.1.1 The Subnetting Formula

The fundamental formula for subnetting is:

```
2^n = number of subnets
```

Where 'n' is the number of bits borrowed from the host portion of the address. Similarly, the number of usable hosts per subnet is calculated as:

```
2^h - 2 = number of usable hosts
```

Where 'h' is the number of remaining host bits. We subtract 2 because the first address (all host bits = 0) is the Network ID and the last address (all host bits = 1) is the Broadcast address.

2.1.2 Our Subnetting Approach

For this project, we started with the Class C network 192.168.40.0/24. The default subnet mask for Class C is 255.255.255.0, which gives us 254 usable hosts in a single network. However, we needed multiple separate subnets.

We chose a /25 subnet mask (255.255.255.128), which means we borrowed 1 bit from the host portion:

Borrowed bits: 25 - 24 = 1 bit

Number of subnets: 2^1 = 2 subnets per /24 network

Host bits remaining: 32 - 25 = 7 bits

Usable hosts per subnet: 2^7 - 2 = 126 hosts

2.1.3 Block Size Calculation

The block size determines how large each subnet is and where the next subnet begins. It is calculated by:

```
Block Size = 256 - subnet mask value in the interesting octet
```

For our /25 mask (255.255.255.128), the interesting octet (4th octet) has a value of 128:

Block Size = 256 - 128 = 128

This means each subnet contains 128 addresses (including Network ID and Broadcast), and subnets start at intervals of 128:

Subnet	Starting Address	Calculation
Subnet 1	192.168.40.0	First block: starts at 0
Subnet 2	192.168.40.128	Second block: 0 + 128 = 128

2.1.4 Detailed Subnet Breakdown — Subnet 1

For Subnet 1 (192.168.40.0/25):

Property	Value	Explanation
Network ID	192.168.40.0	All host bits set to 0 — first address in the block
Subnet Mask	255.255.255.128	/25 — 25 network bits, 7 host bits
First Usable Host	192.168.40.1	Network ID + 1 — assigned to the router (gateway)
Last Usable Host	192.168.40.126	Broadcast - 1 — last assignable address
Broadcast Address	192.168.40.127	All host bits set to 1 — last address in the block
Total Usable Hosts	126	2^7 - 2 = 126

2.1.5 Detailed Subnet Breakdown — Subnet 2

For Subnet 2 (192.168.40.128/25):

Property	Value	Explanation
Network ID	192.168.40.128	Starts at the second block (128)
Subnet Mask	255.255.255.128	/25 — same mask as Subnet 1

First Usable Host	192.168.40.129	Network ID + 1 — assigned to the router (gateway)
Last Usable Host	192.168.40.254	Broadcast - 1
Broadcast Address	192.168.40.255	All host bits set to 1 — end of the block
Total Usable Hosts	126	2^7 - 2 = 126

2.1.6 Expanding Beyond Two Subnets

When the project required a third subnet (for Router2's network), the 192.168.40.0/24 range was already fully allocated — both /25 blocks were in use. Therefore, we moved to 192.168.41.0/25 for Subnet 3, 192.168.42.0/25 for the wireless network (Subnet 4), and 192.168.43.0/25 for the server network (Subnet 5). The /30 network 10.0.0.0/30 was used for the router-to-router link, providing exactly 2 usable addresses.

2.2 Complete Subnet Allocation Table

Subnet	Network ID	Mask	Gateway	Usable Range	Broadcast	Purpose
Subnet 1	192.168.40.0	/25	.1	.1 - .126	.127	Accounts Dept
Subnet 2	192.168.40.128	/25	.129	.129 - .254	.255	Delivery Dept
Subnet 3	192.168.41.0	/25	.1	.1 - .126	.127	General Users (R2)
Subnet 4	192.168.42.0	/25	.1	.1 - .126	.127	Wireless Network
Subnet 5	192.168.43.0	/25	.1	.1 - .126	.127	Server Farm
Router Link	10.0.0.0	/30	N/A	.1 - .2	.3	MainRouter-Router2

3. Device Inventory

3.1 Routers

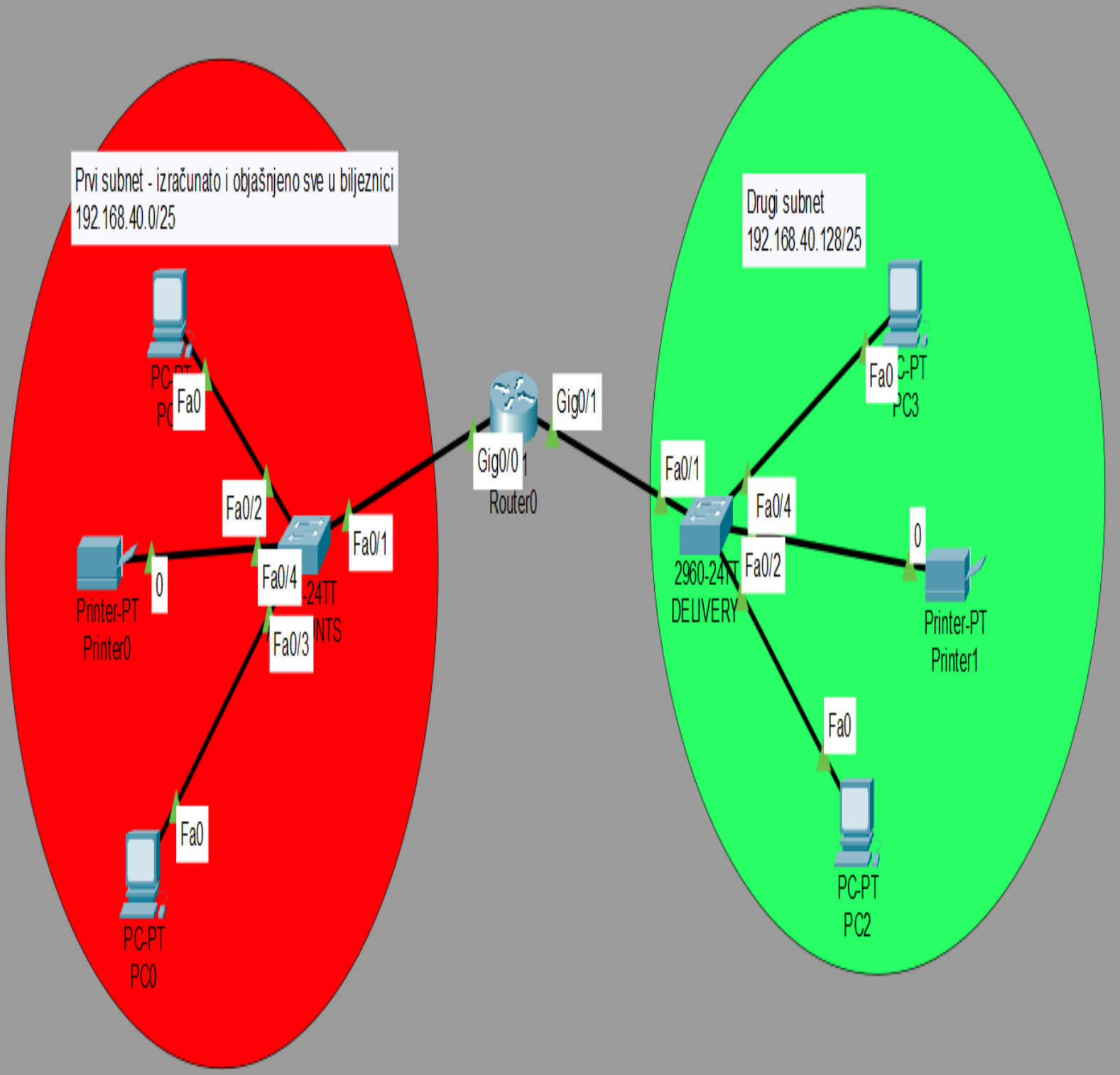
Hostname	Model	Interfaces	Connected To
MainRouter	Cisco 2911	Gig0/0 (.40.1), Gig0/1 (.40.129), Gig0/2 (10.0.0.1), Vlan1 (.42.1)	Switch1, Switch2, Router2, WirelessRouter
Router2	Cisco 2911	Gig0/0 (10.0.0.2), Gig0/1 (.41.1), Gig0/2 (.43.1)	MainRouter, Switch3, ServerSwitch

3.2 Switches

Hostname	Model	Management IP	Subnet	Connected Devices
SwitchAccounts	Cisco 2960	192.168.40.5	Subnet 1	PC0, PC1, Printer1, MainRouter
SwitchDelivery	Cisco 2960	192.168.40.133	Subnet 2	PC2, PC3, Printer2, MainRouter
Switch0	Cisco 2960	N/A	Subnet 3	PC4, PC5, Router2
ServerSwitch	Cisco 2960	N/A	Subnet 5	Server, Router2

3.3 End Devices

Device	Subnet	IP Address	Assignment	Gateway
PC0	Subnet 1	DHCP (.11-.20)	Dynamic	192.168.40.1
PC1	Subnet 1	DHCP (.11-.20)	Dynamic	192.168.40.1
Printer1	Subnet 1	192.168.40.4	Static	192.168.40.1
PC2	Subnet 2	DHCP (.140-.150)	Dynamic	192.168.40.129
PC3	Subnet 2	DHCP (.140-.150)	Dynamic	192.168.40.129
Printer2	Subnet 2	192.168.40.132	Static	192.168.40.129
PC4	Subnet 3	DHCP (.11-.20)	Dynamic	192.168.41.1
PC5	Subnet 3	DHCP (.11-.20)	Dynamic	192.168.41.1
Laptop	Subnet 4	DHCP	Dynamic	192.168.42.1
Server	Subnet 5	192.168.43.10	Static	192.168.43.1


Prvi subnet - izračunato i objašnjeno sve u biljeznici
192.168.40.0/25
Fa0
Fa0/2
Fa0/1
Fa0/4
Fa0/3
0
Printer-PT
Printer0
Fa0
PC-PT
PC0
Gig0/1
Gig0/0
Router0
Drugi subnet
192.168.40.128/25
Fa0
PC3
Fa0/1
Fa0/4
Fa0/2
2960-24TT
DELIVERY
0
Printer-PT
Printer1
Fa0
PC-PT
PC2

4. DHCP Configuration

4.1 Overview

DHCP (Dynamic Host Configuration Protocol) was configured on both routers to automatically assign IP addresses to PCs and the wireless laptop. This eliminates the need for manual IP configuration on each device. The routers serve as DHCP servers because in a small network, it is more efficient to use the existing routers rather than deploying dedicated DHCP servers.

Static devices such as routers, printers, switches, and the server retain manually configured (static) IP addresses. The first 10 addresses in each subnet are reserved for static devices, ensuring clean separation between static and dynamic assignments.

4.2 DHCP Pool Summary

Pool Name	Router	Network	DHCP Range	Excluded Range	DNS Server
SUBNET1	MainRouter	192.168.40.0/25	.11 - .20	.1 - .10	192.168.43.10
SUBNET2	MainRouter	192.168.40.128/25	.140 - .150	.129 - .139	192.168.43.10
WIRELESS1	MainRouter	192.168.42.0/25	.11 - .126	.1 - .10	192.168.43.10
LAN2	Router2	192.168.41.0/25	.11 - .126	.1 - .10	192.168.43.10

4.3 MainRouter DHCP Configuration (CLI)

```
MainRouter(config)# ip dhcp excluded-address 192.168.40.1 192.168.40.10
MainRouter(config)# ip dhcp excluded-address 192.168.40.129 192.168.40.139
MainRouter(config)# ip dhcp excluded-address 192.168.42.1 192.168.42.10

MainRouter(config)# ip dhcp pool SUBNET1
MainRouter(dhcp-config)# network 192.168.40.0 255.255.255.128
MainRouter(dhcp-config)# default-router 192.168.40.1
MainRouter(dhcp-config)# dns-server 192.168.43.10
MainRouter(dhcp-config)# exit

MainRouter(config)# ip dhcp pool SUBNET2
MainRouter(dhcp-config)# network 192.168.40.128 255.255.255.128
MainRouter(dhcp-config)# default-router 192.168.40.129
MainRouter(dhcp-config)# dns-server 192.168.43.10
MainRouter(dhcp-config)# exit

MainRouter(config)# ip dhcp pool WIRELESS1
MainRouter(dhcp-config)# network 192.168.42.0 255.255.255.128
MainRouter(dhcp-config)# default-router 192.168.42.1
MainRouter(dhcp-config)# dns-server 192.168.43.10
```

```
MainRouter(dhcp-config)# exit
```

4.4 Router2 DHCP Configuration (CLI)

```
Router2(config)# ip dhcp excluded-address 192.168.41.1 192.168.41.10

Router2(config)# ip dhcp pool LAN2
Router2(dhcp-config)# network 192.168.41.0 255.255.255.128
Router2(dhcp-config)# default-router 192.168.41.1
Router2(dhcp-config)# dns-server 192.168.43.10
Router2(dhcp-config)# exit
```

4.5 Verifying DHCP

To verify DHCP is functioning correctly, the following commands were used:

```
MainRouter# show ip dhcp pool
MainRouter# show ip dhcp binding
```

On each PC, the IP configuration was switched from Static to DHCP under Desktop > IP Configuration. Each PC successfully received an IP address from the correct pool, along with the correct subnet mask, default gateway, and DNS server address.

5. OSPF Dynamic Routing

5.1 Overview

OSPF (Open Shortest Path First) is a dynamic routing protocol that allows routers to automatically discover and share network routes. Before OSPF was configured, each router only knew about its directly connected networks. PCs in Subnet 3 (behind Router2) could not communicate with PCs in Subnets 1 or 2 (behind MainRouter) because neither router knew how to reach the other's networks.

OSPF solves this by having routers exchange routing information. Once configured, MainRouter automatically learned about Subnet 3 (192.168.41.0/25) and Subnet 5 (192.168.43.0/25) from Router2, and Router2 learned about Subnets 1, 2, and 4 from MainRouter. All networks were placed in OSPF Area 0 (the backbone area).

5.2 OSPF Network Advertisements

Router	Network	Wildcard Mask	Area	Purpose
MainRouter	192.168.40.0	0.0.0.127	0	Subnet 1 (Accounts)
MainRouter	192.168.40.128	0.0.0.127	0	Subnet 2 (Delivery)
MainRouter	10.0.0.0	0.0.0.3	0	Router link
MainRouter	192.168.42.0	0.0.0.127	0	Wireless subnet
Router2	192.168.41.0	0.0.0.127	0	Subnet 3
Router2	192.168.43.0	0.0.0.127	0	Server subnet
Router2	10.0.0.0	0.0.0.3	0	Router link

5.2.1 Understanding Wildcard Masks

OSPF uses wildcard masks instead of subnet masks. The wildcard mask is the inverse of the subnet mask, calculated by subtracting each octet from 255:

Subnet mask 255.255.255.128 becomes wildcard **0.0.0.127** (255 - 128 = 127)

Subnet mask 255.255.255.252 becomes wildcard **0.0.0.3** (255 - 252 = 3)

5.3 MainRouter OSPF Configuration (CLI)

```
MainRouter(config)# router ospf 1
MainRouter(config-router)# network 192.168.40.0 0.0.0.127 area 0
MainRouter(config-router)# network 192.168.40.128 0.0.0.127 area 0
MainRouter(config-router)# network 10.0.0.0 0.0.0.3 area 0
MainRouter(config-router)# network 192.168.42.0 0.0.0.127 area 0
MainRouter(config-router)# exit
```

5.4 Router2 OSPF Configuration (CLI)

```
Router2(config)# router ospf 1
Router2(config-router)# network 192.168.41.0 0.0.0.127 area 0
Router2(config-router)# network 192.168.43.0 0.0.0.127 area 0
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router2(config-router)# exit
```

5.5 Verifying OSPF

The OSPF adjacency was confirmed when the following message appeared in the CLI:

```
%OSPF-5-ADJCHG: Process 1, Nbr 192.168.41.1 on GigabitEthernet0/2
  from LOADING to FULL, Loading Done
```

The routing table on MainRouter showed OSPF-learned routes marked with 'O':

```
MainRouter# show ip route
O    192.168.41.0/25 [110/2] via 10.0.0.2, GigabitEthernet0/2
O    192.168.43.0/25 [110/2] via 10.0.0.2, GigabitEthernet0/2
```

Similarly, Router2 learned about MainRouter's subnets:

```
Router2# show ip route
O    192.168.40.0/25 [110/2] via 10.0.0.1, GigabitEthernet0/0
O    192.168.40.128/25 [110/2] via 10.0.0.1, GigabitEthernet0/0
```

From PC4 (Subnet3) – 192.168.41.12 → PC0 (Subnet1) 192.168.40.12

```
C:\>ping 192.168.40.12

Pinging 192.168.40.12 with 32 bytes of data:

Reply from 192.168.40.12: bytes=32 time<1ms TTL=126
Reply from 192.168.40.12: bytes=32 time<1ms TTL=126
Reply from 192.168.40.12: bytes=32 time<1ms TTL=126
Reply from 192.168.40.12: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.40.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

6. Network Security

6.1 Overview

All network devices were secured with multiple layers of protection: console passwords (physical access), enable secrets (privileged mode access), and SSH with username/password authentication (remote access). Telnet was disabled on all devices, forcing the use of encrypted SSH connections only.

6.2 Security Configuration Summary

Device	Hostname	Enable Secret	Console Password	SSH Username	Domain Name
MainRouter	MainRouter	ADMIN	ADMIN	Patrik	glavniruter.local
Router2	Subnet3Router	ADMIN	subnet3admin	N/A	N/A
Switch 1	SwitchAccounts	switchAcc	switchAcc	SwitchAccAdmin	accounts.local
Switch 2	SwitchDelivery	switchDel	switchDel	SwitchDelAdmin	deliverys.local

6.3 MainRouter Security Configuration (CLI)

```
! Enable secret password
MainRouter(config)# enable secret ADMIN

! Console line security
MainRouter(config)# line console 0
MainRouter(config-line)# password ADMIN
MainRouter(config-line)# login
MainRouter(config-line)# exit

! SSH configuration
MainRouter(config)# hostname MainRouter
MainRouter(config)# ip domain-name glavniruter.local
MainRouter(config)# username Patrik secret ADMIN321
MainRouter(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024

! VTY line configuration (SSH only)
MainRouter(config)# line vty 0 4
MainRouter(config-line)# transport input ssh
MainRouter(config-line)# login local
MainRouter(config-line)# exit
```

6.4 Switch Security Configuration Example (CLI)

The same pattern was applied to all switches. Example for SwitchAccounts:

```
Switch(config)# enable secret switchAcc
Switch(config)# hostname SwitchAccounts
SwitchAccounts(config)# ip domain-name accounts.local
SwitchAccounts(config)# username SwitchAccAdmin secret switchAcc
SwitchAccounts(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024

SwitchAccounts(config)# line console 0
SwitchAccounts(config-line)# password switchAcc
SwitchAccounts(config-line)# login
SwitchAccounts(config-line)# exit

SwitchAccounts(config)# line vty 0 4
SwitchAccounts(config-line)# transport input ssh
SwitchAccounts(config-line)# login local
SwitchAccounts(config-line)# exit
```

6.5 Switch Management IP Addresses

To enable SSH access to switches, management IP addresses were assigned to the Vlan1 interface:

```
SwitchAccounts(config)# interface vlan 1
SwitchAccounts(config-if)# ip address 192.168.40.5 255.255.255.128
SwitchAccounts(config-if)# no shutdown

SwitchDelivery(config)# interface vlan 1
SwitchDelivery(config-if)# ip address 192.168.40.133 255.255.255.128
SwitchDelivery(config-if)# no shutdown
```

6.6 Testing SSH Access

SSH was tested from PCs in each subnet to verify remote management capability:

```
C:\> ssh -l Patrik 192.168.40.1
Password: ****
MainRouter>

C:\> ssh -l SwitchAccAdmin 192.168.40.5
Password: ****
SwitchAccounts>
```

7. VLANs (Virtual LANs)

7.1 Overview

VLANs were configured as a learning exercise to demonstrate traffic segmentation on a single switch. Two VLANs were created on both switches: VLAN 10 for PCs and VLAN 20 for Printers. This demonstrated that devices in different VLANs cannot communicate even though they are connected to the same physical switch.

After demonstrating the concept, VLANs were removed because implementing inter-VLAN routing (router-on-a-stick) would require additional subnets for each VLAN. This was a deliberate design decision to keep the network manageable while still learning the VLAN concept.

7.2 VLAN Configuration (CLI)

```
! Creating VLANs
Switch(config)# vlan 10
Switch(config-vlan)# name PCs
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Printers
Switch(config-vlan)# exit

! Assigning ports to VLANs
Switch(config)# interface Fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface Fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface Fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

7.3 VLAN Verification and Testing

The show vlan brief command confirmed the port assignments:

```
Switch# show vlan brief

VLAN  Name      Status    Ports
----  --------  --------  ----------------------------
1     default   active    Fa0/1, Fa0/5-24, Gig0/1-2
10    PCs       active    Fa0/2, Fa0/3
20    Printers  active    Fa0/4
```

Testing confirmed that PCs in VLAN 10 could ping each other but could NOT ping the printer in VLAN 20, demonstrating successful traffic isolation. Pinging between VLANs returned 'Request timed out' — proving the invisible wall between VLANs was working as expected.

8. Wireless Network Configuration

8.1 Overview

A wireless network was added to allow a laptop to connect to the network without a physical cable. This was accomplished by connecting a WirelessRouter (in Wireless AP mode) to the MainRouter's HWIC-4ESW module port. The wireless network operates on its own subnet (192.168.42.0/25).

8.2 Physical Module Installation

MainRouter required additional ports to support the wireless connection. A HWIC-4ESW (4-port Ethernet switch module) was installed:

1. MainRouter was powered off in Packet Tracer (Physical tab)

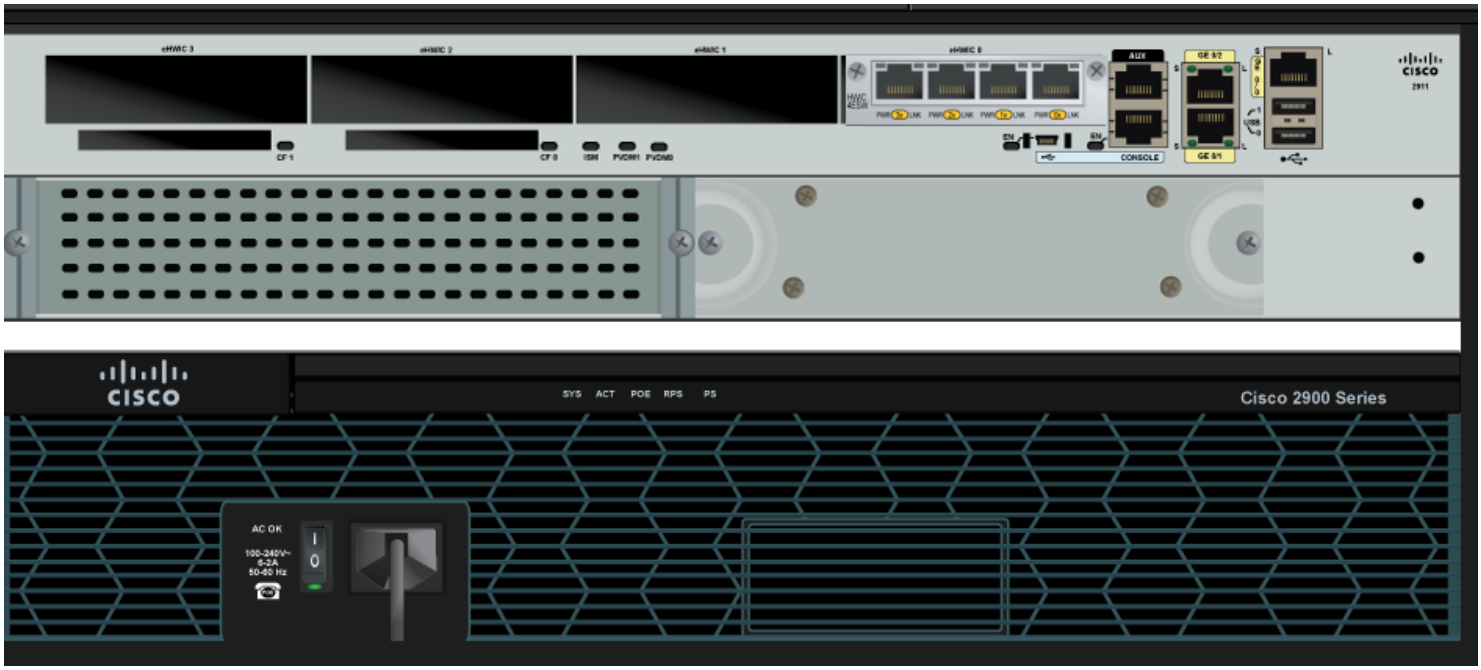
2. The HWIC-4ESW module was dragged into an empty HWIC slot

3. MainRouter was powered back on

4. Four new FastEthernet ports (Fa0/0/0 - Fa0/0/3) appeared

Because the HWIC-4ESW ports function as switch ports (Layer 2), an IP address could not be assigned directly to FastEthernet0/0/0. Instead, the Vlan1 interface was used as the gateway:

```
MainRouter(config)# interface Vlan1
MainRouter(config-if)# ip address 192.168.42.1 255.255.255.128
MainRouter(config-if)# no shutdown
```



8.3 WirelessRouter Configuration

The WirelessRouter was configured via its GUI interface:

Setting	Value
Internet Connection Type	Wireless AP
DHCP Server	Disabled (MainRouter handles DHCP)
SSID (2.4 GHz)	LaptopWIFI
Security Mode	WPA2 Personal
Passphrase	Configured (not shown for security)

8.4 Laptop Configuration

The laptop required a wireless network interface card:

1. The laptop was powered off

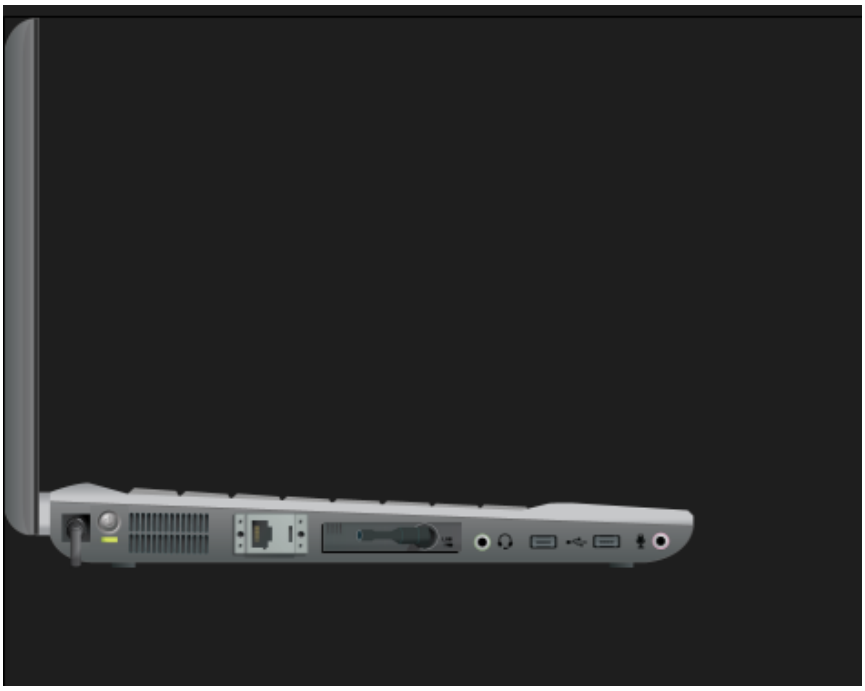
2. The default Ethernet module was removed from the Physical tab

3. A WPC300N wireless module was installed

4. The laptop was powered back on

5. Under Desktop > PC Wireless, the laptop connected to the 'LaptopWIFI' SSID

6. IP configuration was set to DHCP and the laptop received an address from the 192.168.42.0/25 pool



Port Status: On
Bandwidth: 300 Mbps
MAC Address: 0090.2BB5.A033
SSID: LaptopWIFI

Authentication
- Disabled
- WEP
- WPA-PSK
- WPA2-PSK
- WPA
- WPA2
- 802.1X

Method:

WEP Key:
PSK Pass Phrase: ADMIN1234
User ID:
Password:
MD5
User Name:
Password:

Encryption Type: TKIP

IP Configuration
- DHCP
- Static

IPv4 Address: 192.168.42.11
Subnet Mask: 255.255.255.128

IPv6 Configuration
- Automatic
- Static

IPv6 Address:
Link Local Address: FE80::290:2BFF:FEB5:A033

9. Server Services

9.1 Server Overview

A dedicated server was deployed in Subnet 5 (192.168.43.0/25) with a static IP address of 192.168.43.10. The server runs four services: DNS, HTTP (Web), Email (SMTP/POP3), and FTP. This centralized approach simulates a real-world server farm that provides essential network services to all users.

Service	Protocol/Port	Status	Purpose
DNS	UDP 53	ON	Resolve www.copa.hr to 192.168.43.10
HTTP	TCP 80	ON	Host company website
SMTP	TCP 25	ON	Send emails
POP3	TCP 110	ON	Receive emails
FTP	TCP 21	ON	File sharing with role-based access

9.2 DNS Configuration

The DNS service translates the domain name www.copa.hr into the server's IP address (192.168.43.10). This allows users to access the company website by typing a URL instead of remembering an IP address.

On the Server: Services > DNS > ON, with the following A record:

Record Type	Domain Name	IP Address
A Record	www.copa.hr	192.168.43.10

All DHCP pools were configured with the dns-server 192.168.43.10 command so that every device automatically knows where to send DNS queries.

9.3 Web Server (HTTP)

The HTTP service hosts the company website at www.copa.hr. The default index.html page was customized with company content. Any device on the network can open a web browser, type www.copa.hr, and the DNS server resolves it to 192.168.43.10, loading the company website.

[SCREENSHOT: Web browser on a PC showing www.copa.hr loaded successfully]

9.4 Email Server (SMTP/POP3)

The email service allows users across different subnets to send and receive emails using the @copa.hr domain. Both SMTP (sending) and POP3 (receiving) services were enabled on the server.

9.4.1 Email User Accounts

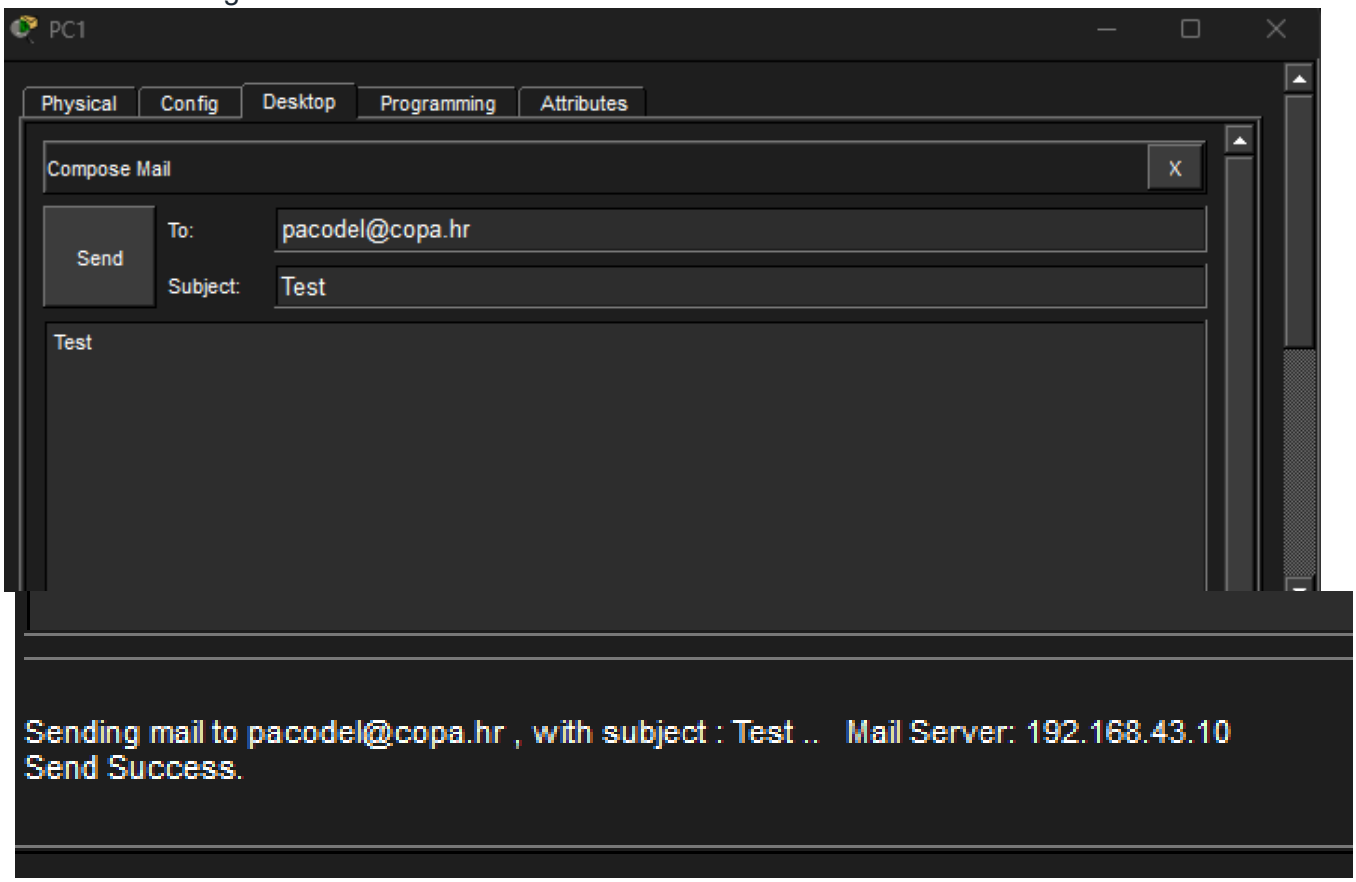
Username	Email Address	Department
pasacc	pasacc@copa.hr	Accounts
luxacc	luxacc@copa.hr	Accounts
pacodel	pacodel@copa.hr	Delivery
vekidel	vekidel@copa.hr	Delivery

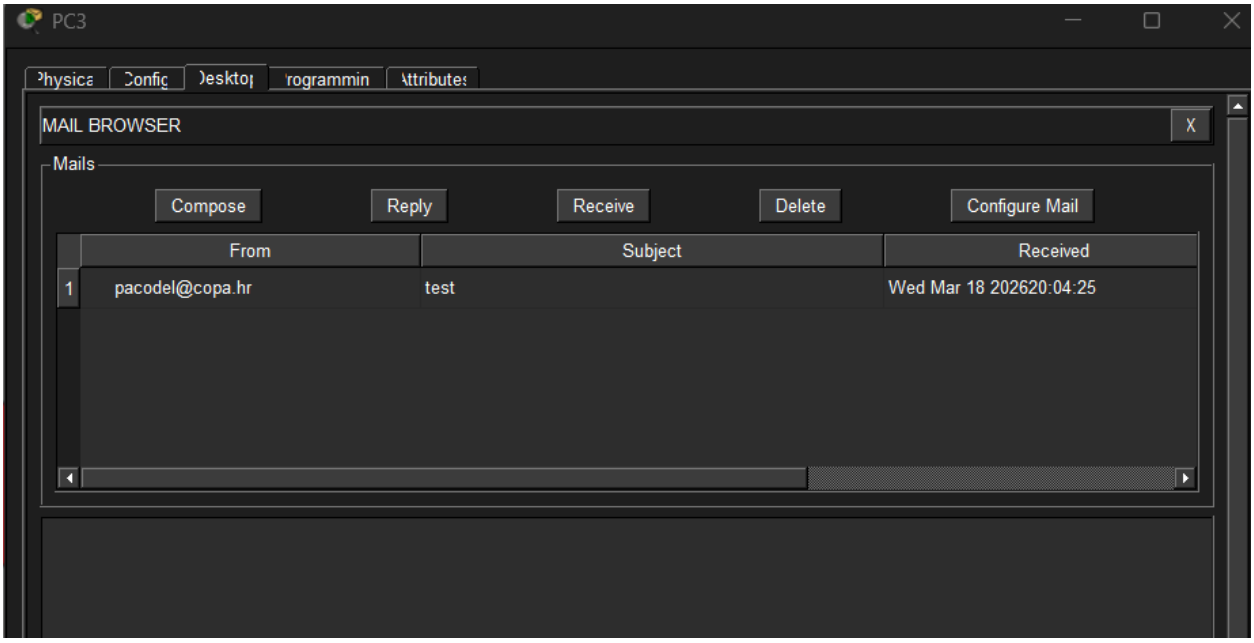
9.4.2 Email Client Configuration on PCs

Each PC was configured under Desktop > Email with the following settings:

Setting	Value
Incoming Mail Server	192.168.43.10
Outgoing Mail Server	192.168.43.10
Username	(matching server account)
Password	(matching server password)

After configuration, emails were successfully sent between users in different subnets, demonstrating that the email service works across the entire OSPF-routed network.





9.5 FTP Server

The FTP service provides file sharing with role-based access control. Different user accounts have different permission levels, simulating a real corporate environment where not all employees should have the same access to files.

9.5.1 FTP User Accounts and Permissions

Username	Permissions	Role Description
admin	RWDNL (Full Access)	Administrator / Manager — can read, write, delete, rename, and list all files
accounts1	RWD	Accounts department — can read, write, and delete files
accounts2	RWD	Accounts department — can read, write, and delete files
delivery1	R (Read Only)	Delivery department — can only download files, cannot modify
delivery2	R (Read Only)	Delivery department — can only download files, cannot modify

9.5.2 FTP Permission Legend

Permission	Letter	Description
Read	R	Download files from the server
Write	W	Upload files to the server
Delete	D	Remove files from the server
Rename	N	Rename files on the server
List	L	View directory contents

9.5.3 FTP Testing

FTP access was tested from the command prompt on PCs:

```
! Testing as delivery1 (Read-only)
C:\> ftp 192.168.43.10
User: delivery1
ftp> dir
%Error: Permission denied    <-- Cannot list (no L permission)

! Testing as accounts1 (RWD)
C:\> ftp 192.168.43.10
User: accounts1
ftp> dir
Listing /ftp directory...    <-- Can list files
ftp> put sample.txt
Transfer complete            <-- Can upload files
```

This demonstrated that the permission system correctly restricts access based on user roles — delivery staff can only read files while accounts staff have broader access.

10. Inter-Router Connectivity

10.1 Point-to-Point Link

MainRouter and Router2 are connected via a dedicated point-to-point link using the 10.0.0.0/30 subnet. A /30 mask was chosen because it provides exactly 2 usable IP addresses — perfect for a link between two routers with no wasted addresses.

Device	Interface	IP Address	Role
MainRouter	GigabitEthernet0/2	10.0.0.1	Near end
Router2	GigabitEthernet0/0	10.0.0.2	Far end

10.2 Configuration (CLI)

```
! MainRouter
MainRouter(config)# interface GigabitEthernet0/2
MainRouter(config-if)# ip address 10.0.0.1 255.255.255.252
MainRouter(config-if)# no shutdown

! Router2
Router2(config)# interface GigabitEthernet0/0
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
```

10.3 Router2 Physical Module

Router2 also required a physical module for additional connectivity. A HWIC-1GE-SFP module with a GLC-LH-SMD SFP transceiver was installed to provide an additional GigabitEthernet port (Gig0/0/0). This module provides a routed port (unlike the HWIC-4ESW which provides switch ports) and was intended for the ISP connection.

Installation process: Power off Router2, insert HWIC-1GE-SFP module into an empty HWIC slot, insert GLC-LH-SMD transceiver into the SFP port, power on Router2.

11. Configuration Management

11.1 Saving Configurations

Cisco devices have two types of configuration: the running-config (active in memory, lost on reboot) and the startup-config (saved to NVRAM, loaded on boot). After making changes, configurations must be saved to survive a reboot:

```
Router# copy running-config startup-config
Destination filename [startup-config]?     <-- Press Enter
Building configuration...
[OK]
```

This command was executed on all routers and switches after configuration changes. During the project, a router reboot without saving caused the loss of OSPF, DHCP, and interface configurations — highlighting the importance of always saving after making changes.

12. Network Testing and Verification

12.1 Connectivity Tests

Comprehensive ping tests were performed from every subnet to verify full network connectivity:

Source	Destination	Result	Path
PC0 (Subnet 1)	PC1 (Subnet 1)	Success	Direct (same switch)
PC0 (Subnet 1)	PC2 (Subnet 2)	Success	Via MainRouter
PC0 (Subnet 1)	PC4 (Subnet 3)	Success	Via MainRouter > Router2
PC0 (Subnet 1)	Server (Subnet 5)	Success	Via MainRouter > Router2
PC4 (Subnet 3)	Printer1 (Subnet 1)	Success	Via Router2 > MainRouter
Laptop (Wireless)	PC2 (Subnet 2)	Success	Via MainRouter
Laptop (Wireless)	Server (Subnet 5)	Success	Via MainRouter > Router2

12.2 Service Tests

Test	Method	Result
DHCP	Set PC to DHCP mode	All PCs received correct IPs from pools
DNS	Browse www.copa.hr	Domain resolved to 192.168.43.10
Web	Open www.copa.hr in browser	Company website displayed
Email	Send email between subnets	Email delivered successfully
FTP Upload	ftp > put sample.txt (as accounts1)	Upload successful
FTP Denied	ftp > dir (as delivery1)	Permission denied (correct)
SSH	ssh -l Patrik 192.168.40.1	Remote CLI access successful

12.3 Complete Traffic Flow Example

The following demonstrates a complete user workflow across the entire network:

Step 1: A user opens the laptop and connects to the 'LaptopWIFI' wireless network using WPA2 authentication.

Step 2: DHCP on MainRouter automatically assigns the laptop an IP address (192.168.42.x), subnet mask, default gateway (192.168.42.1), and DNS server (192.168.43.10).

Step 3: The user opens a web browser and types www.copa.hr.

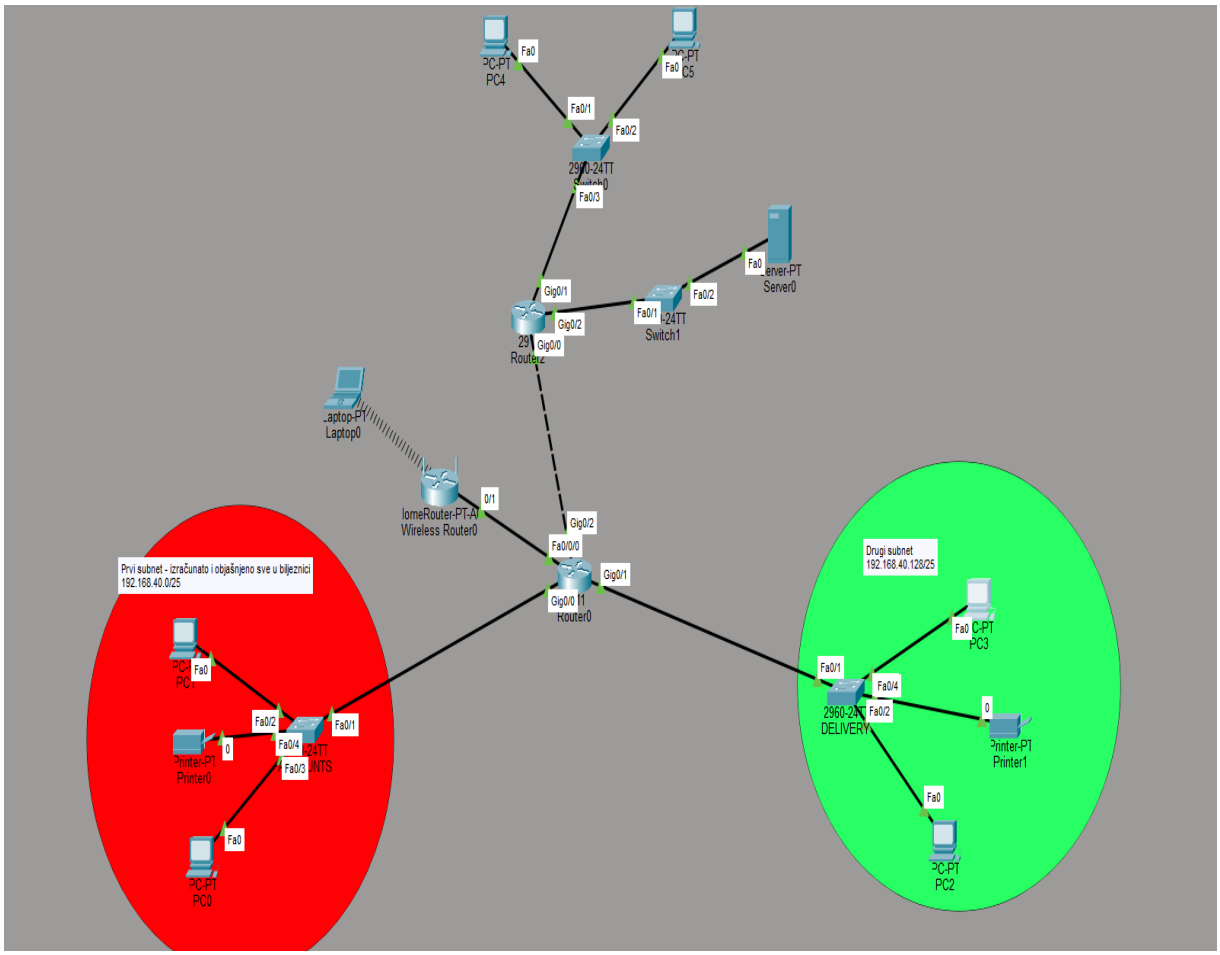
Step 4: The laptop sends a DNS query to 192.168.43.10 (Server in Subnet 5). The query travels from the laptop through the WirelessRouter to MainRouter, then via OSPF-learned route through the 10.0.0.0/30 link to Router2, and finally to the server.

Step 5: The DNS server responds with the IP address 192.168.43.10.

Step 6: The laptop sends an HTTP request to 192.168.43.10, following the same path. The server responds with the company website, and the user sees the Copa.hr homepage.

Step 7: The user then opens the email client and sends an email to a colleague in the Accounts department (Subnet 1). The email travels through MainRouter, to Router2, reaches the email server, and is stored for the recipient to retrieve.

This workflow demonstrates DHCP, DNS, OSPF routing, HTTP, and email services all working together seamlessly across multiple subnets and two routers.



13. Lessons Learned and Troubleshooting

13.1 Key Lessons

VLANs require separate subnets for inter-VLAN routing. When VLANs were configured on the switches, devices in different VLANs could not communicate. Implementing 'router on a stick' for inter-VLAN routing requires each VLAN to have its own subnet and sub-interface on the router. This was attempted but removed to keep the project manageable.

HWIC-4ESW ports are switch ports, not router ports. IP addresses cannot be assigned directly to FastEthernet ports on the HWIC-4ESW module. Instead, the Vlan1 interface must be used. This caused confusion when setting up both the wireless network and NAT, and was a significant learning experience about the difference between Layer 2 and Layer 3 ports.

Always save configurations. When Router2 was rebooted to install a physical module, the running configuration was lost because it had not been saved to startup-config. This resulted in OSPF, DHCP, and interface configurations being erased and needing to be reconfigured. The command 'copy running-config startup-config' should always be run after making changes.

ARP causes first-ping failures. The first ping to a new destination often fails (shown as 80% success rate) because the router needs to perform an ARP request to discover the destination's MAC address. Subsequent pings succeed at 100%.

DHCP changes require lease renewal. After adding the DNS server address to DHCP pools, existing PCs did not receive the updated information until their IP configuration was refreshed (switched from DHCP to Static and back to DHCP).

Default Gateway vs IP Address confusion. When verifying DHCP assignments, it is important to read the correct field. The Default Gateway shows the router's IP, while the IP Address field shows the device's assigned address.

13.2 NAT Attempt

An attempt was made to configure NAT (Network Address Translation) to simulate internet connectivity. An ISP router was added and connected to Router2. However, the HWIC-4ESW module's Vlan1 interface did not support the 'ip nat inside source list' command with the 'overload' option. A HWIC-1GE-SFP module with GLC-LH-SMD SFP transceiver was then tried, but cable compatibility issues in Packet Tracer prevented a successful connection. While the NAT concept was thoroughly understood, the implementation was not completed due to these Packet Tracer hardware limitations.

14. Future Enhancements

The following enhancements could be implemented to further develop this network:

NAT Implementation: Connect the ISP router using a standard GigabitEthernet port to enable private-to-public IP translation and simulate internet access.

ACLs (Access Control Lists): Implement security rules to control which subnets can access specific services. For example, blocking Delivery department PCs from accessing the FTP server's write functions at the network level.

VLANs with Proper Subnetting: Redesign the IP addressing scheme to allocate separate subnets for each VLAN, then implement full inter-VLAN routing using the router-on-a-stick method.

Third Router with OSPF: Add a third router to demonstrate OSPF scalability and multi-path routing.

NTP (Network Time Protocol): Synchronize time across all network devices for consistent logging.

Syslog Server: Centralize device logs for monitoring and troubleshooting.

15. Verification Commands Reference

15.1 Overview

The following show commands were used throughout the project to verify configurations and troubleshoot issues. For each command, a description is provided along with an indication of where to capture a screenshot for documentation purposes.

15.2 Router Verification Commands

15.2.1 show ip interface brief

Displays a summary of all interfaces, their IP addresses, and whether they are up or down. This is the most frequently used verification command.

```
MainRouter# show ip interface brief
Interface              IP-Address      OK? Method Status    Protocol
GigabitEthernet0/0     192.168.40.1    YES NVRAM  up        up
GigabitEthernet0/1     192.168.40.129  YES NVRAM  up        up
GigabitEthernet0/2     10.0.0.1        YES manual up        up
FastEthernet0/0/0      unassigned      YES unset  up        up
Vlan1                  192.168.42.1    YES manual up        up
```

MainRouter:

```
MainRouter#show ip interface br
MainRouter#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.40.1    YES NVRAM  up                    up
GigabitEthernet0/1     192.168.40.129  YES NVRAM  up                    up
GigabitEthernet0/2     10.0.0.1        YES manual up                    up
FastEthernet0/0/0      unassigned      YES unset  up                    up
FastEthernet0/0/1      unassigned      YES unset  down                  down
FastEthernet0/0/2      unassigned      YES unset  down                  down
FastEthernet0/0/3      unassigned      YES unset  down                  down
Vlan1                  192.168.42.1    YES manual up                    up
MainRouter#
```

Router2:

```
Router2>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.2        YES NVRAM  up                    up
GigabitEthernet0/1     192.168.41.1    YES NVRAM  up                    up
GigabitEthernet0/2     192.168.43.1    YES NVRAM  up                    up
Vlan1                  209.165.200.225 YES NVRAM  up                    down
Router2>
```

15.2.2 show ip route

Displays the routing table showing all known routes. Routes marked with 'C' are directly connected, 'L' are local, and 'O' are learned via OSPF. This command proves that OSPF is sharing routes between routers.

```
MainRouter# show ip route
...
C    192.168.40.0/25 is directly connected, GigabitEthernet0/0
C    192.168.40.128/25 is directly connected, GigabitEthernet0/1
C    10.0.0.0/30 is directly connected, GigabitEthernet0/2
O    192.168.41.0/25 [110/2] via 10.0.0.2, GigabitEthernet0/2
O    192.168.43.0/25 [110/2] via 10.0.0.2, GigabitEthernet0/2
```

```
MainRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
     192.168.40.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.40.0/25 is directly connected, GigabitEthernet0/0
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0
C       192.168.40.128/25 is directly connected, GigabitEthernet0/1
L       192.168.40.129/32 is directly connected, GigabitEthernet0/1
     192.168.41.0/25 is subnetted, 1 subnets
O       192.168.41.0/25 [110/2] via 10.0.0.2, 01:15:13, GigabitEthernet0/2
     192.168.42.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.42.0/25 is directly connected, Vlan1
L       192.168.42.1/32 is directly connected, Vlan1
     192.168.43.0/25 is subnetted, 1 subnets
O       192.168.43.0/25 [110/2] via 10.0.0.2, 01:15:13, GigabitEthernet0/2

MainRouter#
```

```
Router2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
     192.168.40.0/25 is subnetted, 2 subnets
O       192.168.40.0/25 [110/2] via 10.0.0.1, 01:16:23, GigabitEthernet0/0
O       192.168.40.128/25 [110/2] via 10.0.0.1, 01:16:23, GigabitEthernet0/0
     192.168.41.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.41.0/25 is directly connected, GigabitEthernet0/1
L       192.168.41.1/32 is directly connected, GigabitEthernet0/1
     192.168.42.0/25 is subnetted, 1 subnets
O       192.168.42.0/25 [110/2] via 10.0.0.1, 01:16:23, GigabitEthernet0/0
     192.168.43.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.43.0/25 is directly connected, GigabitEthernet0/2
L       192.168.43.1/32 is directly connected, GigabitEthernet0/2
```

15.2.3 show ip ospf neighbor

Displays OSPF neighbor adjacencies. A state of 'FULL' means the routers have successfully exchanged routing information. This confirms OSPF is functioning correctly between the two routers.

```
MainRouter# show ip ospf neighbor

Neighbor ID   Pri  State     Dead Time  Address    Interface
192.168.41.1  1    FULL/DR   00:00:31   10.0.0.2   GigabitEthernet0/2
```

```
MainRouter#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
209.165.200.225   1   FULL/DR         00:00:38    10.0.0.2        GigabitEthernet0/2
MainRouter#
```

15.2.4 show ip ospf interface

Displays detailed OSPF information for each interface, including the area, process ID, network type, hello/dead timers, and neighbor count. Useful for verifying which interfaces are participating in OSPF.

15.2.5 show ip dhcp pool

Displays DHCP pool statistics including total addresses, leased addresses, and excluded addresses. This verifies that DHCP pools are correctly configured and actively handing out addresses.

```
MainRouter# show ip dhcp pool

Pool SUBNET1:
  Total addresses     : 126
  Leased addresses    : 2
  Excluded addresses  : 3

Pool SUBNET2:
  Total addresses     : 126
  Leased addresses    : 2
```

```
Excluded addresses  : 3
MainRouter#show ip dhcp pool

Pool WIRELESS1 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.42.1         192.168.42.1     - 192.168.42.126    1    / 3     / 126

Pool SUBNET1 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.40.1         192.168.40.1     - 192.168.40.126    2    / 3     / 126

Pool SUBNET2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.40.129       192.168.40.129   - 192.168.40.254    2    / 3     / 126
MainRouter#

Router2>show ip dhcp pool

Pool LAN2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 0
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.41.1         192.168.41.1     - 192.168.41.126    0    / 2     / 126
Router2>
```

15.2.6 show ip dhcp binding

Displays a table of all IP addresses that have been dynamically assigned by DHCP, along with the MAC address (Client-ID) of each device. This proves that specific PCs have received addresses from the correct pools.

```
MainRouter# show ip dhcp binding

IP address       Client-ID/              Lease expiration  Type
                 Hardware address
192.168.40.11    00E0.F70D.9A73          --                Automatic
192.168.40.12    0002.1617.AB5B          --                Automatic
```

```
MainRouter#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.42.11    0090.2BB5.A033           --                     Automatic
192.168.40.11    0040.0B59.6356           --                     Automatic
192.168.40.12    00E0.B004.C02D           --                     Automatic
192.168.40.130   0002.1617.AB5B           --                     Automatic
192.168.40.131   00E0.F70D.9A73           --                     Automatic
```

15.2.7 show running-config

Displays the complete current configuration of the device. This is useful for verifying all settings at once and for backing up configurations. Due to length, it is recommended to screenshot specific sections rather than the entire output.

```
MainRouter#show running-config
Building configuration...

Current configuration : 1673 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname MainRouter
!
!
!
!
ip dhcp excluded-address 192.168.42.1 192.168.42.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.41.1 192.168.41.10
!
ip dhcp pool WIRELESS1
 network 192.168.42.0 255.255.255.128
 default-router 192.168.42.1
 dns-server 192.168.43.10
ip dhcp pool SUBNET1
 network 192.168.40.0 255.255.255.128
 default-router 192.168.40.1
 dns-server 192.168.43.10
ip dhcp pool SUBNET2
 network 192.168.40.128 255.255.255.128
 default-router 192.168.40.129
 dns-server 192.168.43.10
```

15.3 Switch Verification Commands

15.3.1 show vlan brief

Displays all VLANs configured on the switch and which ports are assigned to each VLAN. Even though VLANs were later removed, this command was critical during the VLAN learning phase.

```
SwitchAccounts# show vlan brief

VLAN  Name      Status    Ports
----  --------  --------  ---------------------------
1     default   active    Fa0/1, Fa0/2, Fa0/3, Fa0/4...
10    PCs       active    (no ports – VLANs removed)
20    Printers  active    (no ports – VLANs removed)
```

```
SwitchAccounts>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   PCs                              active
20   Printers                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SwitchAccounts>
```

```
SwitchDelivery>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   PCs                              active
20   Printers                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SwitchDelivery>
```

15.3.2 show interfaces trunk

Displays which ports are configured as trunk ports and which VLANs they carry. This was used to verify trunk configuration when attempting inter-VLAN routing.

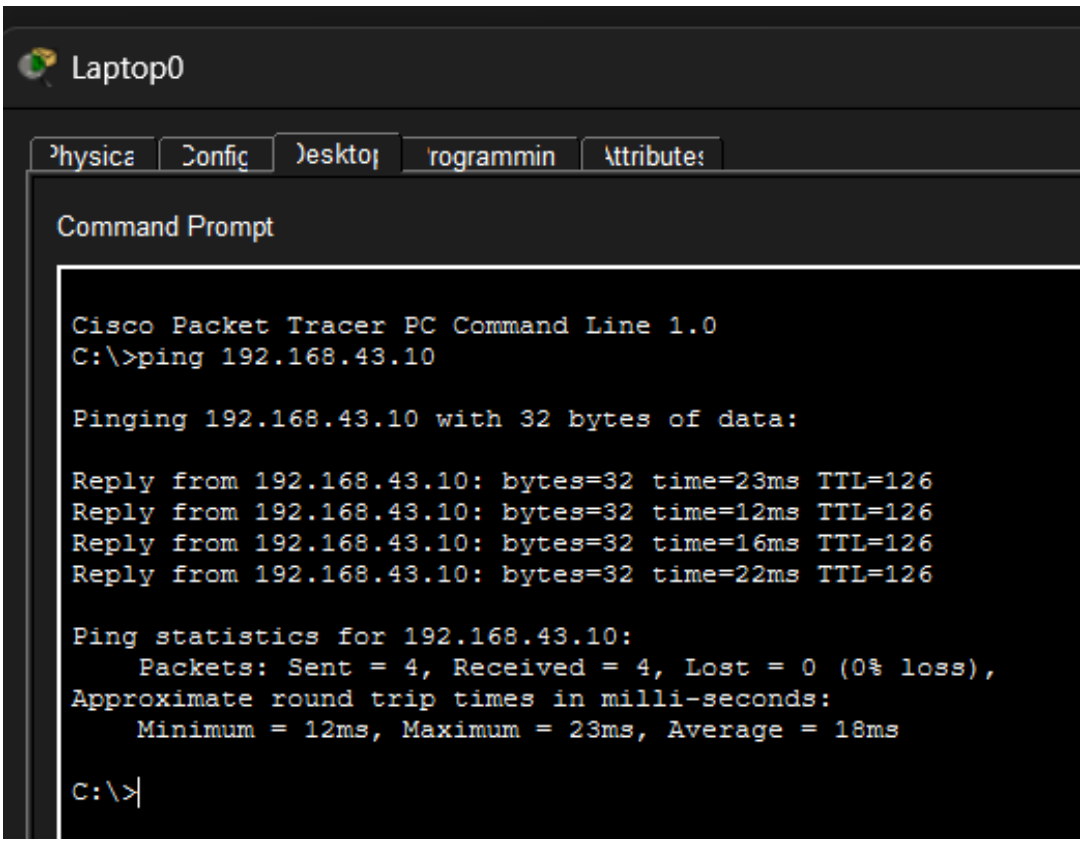
15.3.3 show interfaces status

Displays the status of all switch ports including speed, duplex, and VLAN assignment. Useful for a quick overview of the switch's port usage.

15.4 PC Verification Commands

15.4.1 ping

The ping command tests connectivity between two devices by sending ICMP echo requests. It was used extensively to verify communication within subnets, across subnets (via OSPF), and to the server.



15.4.2 ipconfig

Displays the PC's IP configuration including IP address, subnet mask, default gateway, and DNS server. Used to verify DHCP assignments.

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.9633.CA24
   Link-local IPv6 Address.........: FE80::201:96FF:FE33:CA24
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.41.12
   Subnet Mask.....................: 255.255.255.128
   Default Gateway.................: ::
                                     192.168.41.1
   DHCP Servers....................: 192.168.41.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-0C-9D-ED-3D-00-01-96-33-CA-24
   DNS Servers.....................: ::
                                     192.168.43.10

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 00D0.FFEA.736C
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-0C-9D-ED-3D-00-01-96-33-CA-24
   DNS Servers.....................: ::
                                     192.168.43.10
```

15.4.3 ftp

The FTP command connects to the file server for testing file transfer permissions.

```
C:\>ftp 192.168.43.10
Trying to connect...192.168.43.10
Connected to 192.168.43.10
220- Welcome to PT Ftp server
Username:accounts1
331- Username ok, need password
Password:**********
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.43.10:
0   : asa842-k8.bin                                    5571584
1   : asa923-k8.bin                                    30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3   : c1841-ipbase-mz.123-14.T7.bin                    13832032
4   : c1841-ipbasek9-mz.124-12.bin                     16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7   : c2600-i-mz.122-28.bin                            5571584
8   : c2600-ipbasek9-mz.124-8.bin                      13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                    15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin           33591768
14  : c2950-i6q412-mz.121-22.EA4.bin                   3058048
15  : c2950-i6q412-mz.121-22.EA8.bin                   3117390
16  : c2960-lanbase-mz.122-25.FX.bin                   4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin                 4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin                 4670455
```

15.4.4 ssh

The SSH command establishes an encrypted remote connection to a router or switch.

16. Packet Tracer Screenshots Guide

16.1 Topology Screenshots

The following screenshots should be captured from the Packet Tracer workspace to visually document the network at different stages of the project.

Screenshot	Description	Where to Insert
Initial 2-Subnet Topology	The original network with MainRouter, 2 switches, 4 PCs, and 2 printers	Section 3 (Device Inventory)
After Adding Subnet 3	Network expanded with Router2, Switch0, PC4, and PC5	Section 5 (OSPF)
After Adding Wireless	Network with WirelessRouter and Laptop connected	Section 8 (Wireless)

After Adding Server	Network with Server and ServerSwitch in Subnet 5	Section 9 (Server Services)
Final Complete Topology	Full network showing all devices, subnets, and connections	Section 12 (Testing)

16.2 Service GUI Screenshots

These screenshots should be captured from the Server's Services tab in Packet Tracer.

Screenshot	How to Capture	Where to Insert
DNS Service	Server > Services > DNS — show the A record for www.copa.hr	Section 9.2
HTTP Service	Server > Services > HTTP — show the service is ON	Section 9.3
Email Service	Server > Services > EMAIL — show SMTP/POP3 ON and user accounts	Section 9.4
FTP Service	Server > Services > FTP — show user accounts and permissions	Section 9.5
Email Client Config	Any PC > Desktop > Email — show the configured email settings	Section 9.4
Web Browser	Any PC > Desktop > Web Browser — show www.copa.hr loaded	Section 9.3
DHCP on PC	Any PC > Desktop > IP Configuration — show DHCP-assigned settings	Section 4

16.3 Physical Module Screenshots

These screenshots show the hardware modules installed in the routers.

Screenshot	How to Capture	Where to Insert
MainRouter HWIC-4ESW	MainRouter > Physical tab — show the module in the slot	Section 8.2
Router2 HWIC-1GE-SFP	Router2 > Physical tab — show the module with GLC-LH-SMD	Section 10.3

16.4 Wireless Configuration Screenshots

Screenshot	How to Capture	Where to Insert

WirelessRouter Setup	WirelessRouter > GUI > Setup tab — show Wireless AP mode	Section 8.3
Wireless SSID Config	WirelessRouter > GUI > Wireless tab — show SSID 'LaptopWIFI'	Section 8.3
Wireless Security	WirelessRouter > GUI > Wireless Security — show WPA2 Personal	Section 8.3
Laptop Wireless Connection	Laptop > Desktop > PC Wireless — show connected to LaptopWIFI	Section 8.4

16.5 Security Screenshots

Screenshot	How to Capture	Where to Insert
Console Password Prompt	Close and reopen router CLI — capture the password prompt	Section 6.3
Enable Password Prompt	Type 'enable' from Router> — capture the password prompt	Section 6.3
SSH Login	From PC command prompt: ssh -l Patrik 192.168.40.1	Section 6.6
SSH to Switch	From PC: ssh -l SwitchAccAdmin 192.168.40.5	Section 6.6